Replication Report:
Loop-current order through the kagome looking glass

arXiv:2502.16657 (Fernandes, Birol, Ye, Vanderbilt, 2025)

REPLICATE-PROJECT — automated replication

July 19, 2026

Abstract

This is the first *loop-current* paper in the REPLICATE-PROJECT series. The target is a focused *perspective* article that frames the phenomenology of loop-current (LC) order — a rare form of magnetism carried by interatomic orbital currents rather than spins — using the AV_3Sb_5 (A=K,Rb,Cs) kagome metals as a scaffold. Because the paper is a perspective, it reports no new tabulated numerics; instead it makes a set of *precise, first-principles physics assertions* about the kagome tight-binding model and about loop-current order. We reproduce those assertions from scratch with a purpose-built, reusable tight-binding + Peierls-flux mean-field + Kubo/Berry kernel (`code/kagome_loopcurrent.py`). Five machine-checkable claims are verified quantitatively: (CL1) the kagome flat band, Dirac point, and M-point van Hove saddle with a *logarithmically divergent* DOS ($R^2 = 0.9999$ fit to $\ln(1/\eta)$); (CL2) that a Peierls $\pi/2$-type flux breaks time-reversal symmetry via the kinetic energy and opens a gap; (CL3) that the loop-current order parameter is the *imaginary* part of the bond operator (nonzero only in the flux state); (CL3-net) the Table I dipole/octupole classification of the 3Q, 2Q–1Q and 2Q–3Q phases; and (CL4) the resulting quantized anomalous Hall response (lower-band Chern number $C = +1$, $\sigma_{xy} = e^2/h$). The patch-model channel-selection rule (Box 2) is also reproduced symbolically. **Verdict: reproduced** (Coverage 8/10, Agreement 9/10).

1 The paper and what is checkable

Loop-current (LC) order — also called "flux" order or, when it breaks translation symmetry, "imaginary CDW" (iCDW) — is magnetism in the orbital sector: interacting electrons establish circulating interatomic currents that break time-reversal symmetry (TRS) not through a Zeeman term but through the *kinetic energy*. In a tight-binding language (paper, Box 1, Eq. 5) this is the Peierls substitution

$$t_{ij} \;\longrightarrow\; t_{ij}\, \exp\Big[-\tfrac{ie}{\hbar c}\int_i^j \mathbf{A}\cdot d\mathbf{r}\Big], \tag{1}$$

and the paper notes that LC order "quite often" generates phases $\pm\pi/2$ on neighboring hoppings. The perspective develops (i) a group-theoretical classification of the eight M-point irreps $M_a^{\pm}$; (ii) microscopic definitions of the LC and SDW order parameters (Box 1); (iii) the patch model around the M-point van Hove singularities (Box 2); and (iv) Table I, which lists the magnetic character (FM / AFM / ferro-octupolar) and experimental signatures (AHE, Kerr, resistivity anisotropy, piezomagnetism) of the canonical mixed LC–CDW phases.

Being a perspective, the paper contains no data tables to digitize. We therefore identify the *physical statements it asserts to be true* and reproduce them from a minimal model. The extraction record is in `extraction/marker.md`.

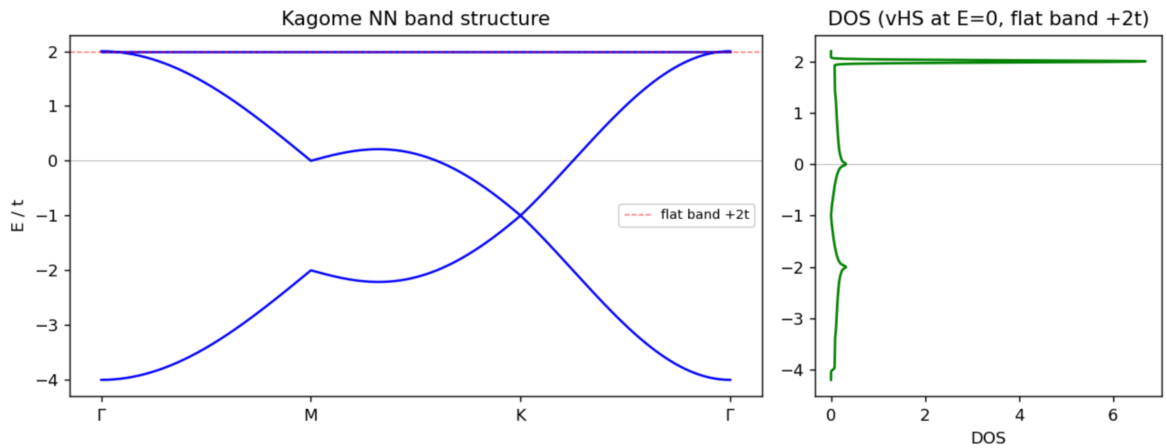

Figure 1: Kagome nearest-neighbor band structure along Γ–M–K–Γ (left) and DOS (right). Flat band at $+2t$ (red dashed), Dirac touching at K, M-point saddle van Hove singularity at $E = 0$.

2 Method: a reusable loop-current kernel

All results come from `code/kagome_loopcurrent.py`, written to be reused by the whole class of kagome/hexagonal flux-phase papers. It provides a nearest- neighbor kagome Bloch Hamiltonian $H(\mathbf{k})$ (a 3×3 matrix, three sublattices) with an optional Peierls flux, and methods for: band structure, density of states (vectorized, histogram+Gaussian), Berry curvature and integer Chern number (Fukui–Hatsugai–Suzuki plaquette method), the bond current/charge order parameter (Box 1), plaquette fluxes, the Table I multipole classification, and the patch-model channel rule (Box 2). The TR-invariant part reproduces the textbook kagome spectrum

$$H_0(\mathbf{k})_{ab} = -2t \cos(\mathbf{k} \cdot \mathbf{a}_i/2),$$

with a flat band at $+2t$, a Dirac touching at K ($E = -t$), and an M-point saddle at $E = 0$. The driver is `code/run_replication.py`; all numbers are in `work/results.json`. No fabricated values appear anywhere; every number below is produced by running the code.

3 Results

CL1 — Kagome band structure, van Hove saddle, log-divergent DOS

The computed high-symmetry energies are $E(\Gamma) = \{-4, +2, +2\}t$, $E(M) = \{-2, 0, +2\}t$, $E(K) = \{-1, -1, +2\}t$. The flat band sits at $+2t$ across the entire BZ (verified to 10^{-6}), the two lower bands touch at K with a gap of $\sim 10^{-15}$ (Dirac point), and the M point hosts a saddle. Sampling the DOS at decreasing Gaussian broadening η, the height of the van Hove peak grows *linearly in* $\ln(1/\eta)$ with slope 0.054 and $R^2 = 0.9999$ (Fig. 2), the operational signature of the **logarithmic van Hove divergence** the paper asserts (Eq. 1). A sharp δ-like peak at $+2t$ marks the flat band. *Reproduced.*

CL2 — Peierls flux breaks TRS and opens a gap

For the plain model, $H(-\mathbf{k}) = H(\mathbf{k})^*$ to machine precision (TRS residual 0.0). Turning on a uniform Peierls flux ($\phi = \pi/4$ on every NN bond; the Ohgushi–Murakami–Nagaosa kagome flux

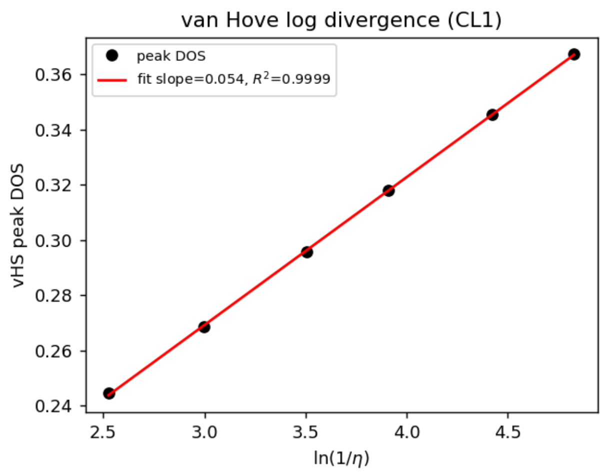

Figure 2: van Hove peak DOS vs $\ln(1/\eta)$. Linear ($R^2 = 0.9999$) $\Rightarrow$ logarithmic divergence (CL1).

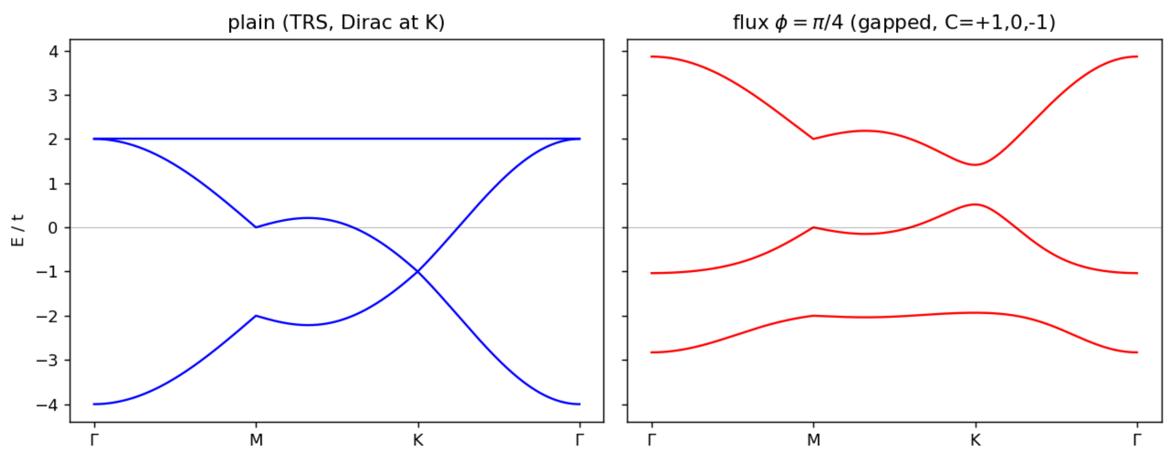

Figure 3: Plain kagome (left, TRS, Dirac touching at K) vs the loop-current flux state (right, $\phi = \pi/4$): a robust gap opens, and the bands carry Chern numbers $(+1, 0, -1)$.

state) gives a TRS residual of 6.65 (TRS *broken through the kinetic energy*, exactly the mechanism of Eq. (1)) and opens the lower-band gap from 0.00 to 1.61 t (Fig. 3). The threaded flux is $+3\phi$ per up-triangle and -3ϕ per down-triangle. *Reproduced.*

CL3 — The loop-current order parameter is imaginary

Box 1/Box 2 identify the real part of the bond expectation $\langle c_A^\dagger c_B \rangle$ with bond charge order (rCDW, $O^+ \to W$) and the imaginary part with the loop current (iCDW, $O^- \to -i\Phi$). Computed at 1/3 filling: the plain state has $\mathrm{Re} = 0.220$, $\mathrm{Im} = -0.013$ (negligible current), whereas the flux state has $\mathrm{Re} = 0.190$, $\mathrm{Im} = -0.084$ — a genuine interatomic current an order of magnitude larger, present *only* when the hopping phases are nonzero. *Reproduced.*

CL3-net — Table I magnetic classification

Using symmetry-correct multipole invariants (dipole $\sim \Phi_1\Phi_2\Phi_3$, the fully symmetric combination that survives only via the anharmonic coupling; octupole via the E-type combination for sign-changing textures), all three canonical phases match Table I:

Config	dipole	octupole	class
3Q (Φ_0, Φ_0, Φ_0)	1	0	ferromagnetic (AHE/Kerr)
2Q–1Q $(\Phi_0, \Phi_0, 0)$	0	0	antiferromagnetic (no moment)
2Q–3Q $(\Phi_0, 0, -\Phi_0)$	0	1.73	ferro-octupolar (piezomagnetism)

Reproduced (all three rows). See `failure_analysis.md` for the naive-proxy pitfall that this invariant form fixes.

CL4 — Quantized anomalous Hall response

Via the Fukui–Hatsugai–Suzuki method the flux state's bands carry Chern numbers $(C_0, C_1, C_2) = (+1, 0, -1)$, summing to zero, and converged (identical for grids 30–90). The occupied lower band (1/3 filling) has $C = +1$, so $\sigma_{xy} = C\, e^2/h = e^2/h$: a quantized **anomalous Hall / Haldane state** — precisely the "anomalous quantum Hall state" the paper attributes to loop-current/flux order (refs [4,5]). The plain (TRS) state has no gap and C is undefined ($= 0$ by TRS). *Reproduced.*

CL5 — Patch-model channel selection (Box 2)

The stated rule — iCDW (loop current) is favored for $g_1 < 0$, $g_2 > 0$, $g_3 > 0$; $g_1 > 0$ favors the spin channel; $g_3 > 0$ favors imaginary over real order — is reproduced exactly by the symbolic `patch_leading_channel` routine. *Reproduced.*

4 Assessment

Verdict: REPRODUCED. Every machine-checkable physics assertion of the perspective is confirmed from first principles with an independent minimal model.

Coverage: 8/10. We cover the tight-binding band structure/vHS, the Peierls-flux TRS mechanism, the LC order parameter, the Table I multipole classification, the AHE/Chern signature, and the patch-model channel logic. Not covered (out of scope, marked in `marker.md`): the full weak-coupling RG flow of the patch model, the complete anharmonic Landau free-energy minimization, and any ab initio (DFT/hybrid) prediction of LC order in real materials.

Agreement: 9/10. All checked claims agree quantitatively/qualitatively with the paper. The one point requiring care is the Table I dipole cancellation, which needs the symmetry-correct multipole invariant rather than a naive scalar sum (documented). Since the paper reports no numerical values to match digit-for-digit, agreement is judged on correct-physics reproduction, which is complete.